Predicting Readmission Rate for Diabetic Patients Using Machine Learning

Team Members and Emails
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Abstract
Hospital readmission rate among diabetes patients is a persistent challenge in the Healthcare industry, highlighting gaps in discharge planning and the difficulty of balancing rising hospital costs with maintaining treatment efficiency. High readmission rates not only signal potential lapses in treatment quality but also strain healthcare resources, underscoring the value of early risk identification. The Centers for Medicare & Medicaid Services (CMS) heightened the urgency to reduce readmissions through its Hospital Readmissions Reduction Program (HRRP) in 2012, imposing financial penalties and public reporting requirements on hospitals with excessive 30-day readmission rates. To address these challenges, this paper utilizes machine learning techniques, including logistic regression, Random Forest, XGBoost, and a Transformer-based neural network to predict 30-day readmission risk using structured electronic health record (EHRs) data from 1999-2008 with the goals of supporting cost reduction for hospitals, improving compliance with CMS regulations, and uncovering which patient demographics (e.g., weight, height, age), dosage changes, and other hospital-related features strongly contribute to the likelihood of readmission. Across all models, tree-based and boosting methods outperformed the logistic regression baseline based on F1 scores, with XGBoost achieving the strongest overall discriminative performance (F1 Score: 57.91%, Test Accuracy: 56.47%). Feature contribution analysis highlighted that the prior utilization of healthcare (*number_inpatient*) and resource utilization during the stay (*time_in_hospital*) were the most influential predictors.

Introduction
Hospital readmission rates are a major concern for both healthcare providers and patients, as they reflect the balance between reducing hospital costs and maintaining treatment efficiency and effectiveness. Among chronic conditions, diabetes is particularly challenging with complicated treatments and high readmission rates. Insufficient follow-up care, partially due to arbitrary diabetes management in hospital environments, and complex discharge procedures have been identified as potential causes of this issue. Given these challenges, predicting which diabetes patients are likely to be readmitted soon after discharge has become a significant clinical focus and research priority. To address this challenge, hospitals often target high-risk diabetes patients with interventions such as patient education programs, medication adjustments, preventive and therapeutic interventions, or closer follow-up appointments to improve care quality and reduce preventable readmissions.

The Readmission rate issue has also escalated to the level of national policy makers, further incentivizing hospitals to reduce readmission rates. For example, in 2012, the Centers for Medicare and Medicaid Services (CMS) launched the Hospital Readmissions Reduction Program (HRRP) which imposed financial penalties and public reporting requirements on hospitals with excessive 30-day readmission rates. Given the persistent relevance, this policy continues to be revised including several revisions between 2020 and 2025 to account for patient case mix, socioeconomic status, and peer-group hospital comparisons, allowing for more equitable penalty calculations and reducing the unfair burden on hospitals serving disadvantaged populations. Although the dataset used in this project predates the introduction of this policy and its revisions, the results remain highly relevant for studying readmission risk given the clinical and demographic factors studied, which drive readmissions, have remained consistent over the last decade. The underlying relationships between patient characteristics, hospital utilization, and readmission likelihood provide insights that generalize beyond the HRRP's context, and support the broader development of predictive models used to inform current clinical interventions.

To identify patients at high risk of complications before discharge, we utilize machine learning models that take structured electronic health record data as inputs including: demographics, hospitalization characteristics, medical dosage changes, and diabetes-related treatment variables. Using models such as logistic regression, Random Forest, XGBoost, and Transformer-based Neural Networks, we predict whether a patient is likely to be readmitted within 30 days of discharge, enabling hospitals to intervene proactively and improve patient outcomes.

Related Work
Research into Hospital readmission prediction among patients with diabetes has increasingly leveraged electronic health records (EHRs) and machine learning models to identify high-risk individuals early in the care process.

Futoma et al. (2015) conducted one of the earliest ML comparisons for diabetes-related readmission using several years of structured inpatient EHR data, including demographic variables, diagnosis codes, and medication changes. The authors utilized logistic regression, support vector machines, Random Forest, and naive Bayes to capture non-linear relationships, concluding that tree-based models outperformed linear methods by capturing high-dimensional interactions and heterogeneous patient subgroups. This work provided an excellent benchmark for our work to build upon and as we adopted deeper representation learning and improved handling of sparse, high-cardinality clinical variables.

Yu et al. (2015) examined diabetes readmission prediction with a focus on the role of medication changes and comorbidity patterns as major predictive signals. Their dataset included diabetes medication regimens, chronic disease indicators, and dosage changes, and they utilized feature engineering to derive variables reflecting the instability in diabetes management, such as dosage changes and treatment intensification. Their experiments showed that models incorporating treatment-level engineered variables substantially outperformed models relying solely on demographics or static clinical indicators. Their emphasis on clinically interpretable predictors helped identify the importance of dynamic treatment signals as predictors of readmission. Our analysis also drew from this insight by placing an emphasis on dosage-related features and hospitalization-related features, which we evaluated under various machine learning frameworks to compare stability and importance.

Hebert et al. (2018) focused on general medical readmissions using the development of gradient-boosted decision tree models for general medical readmissions and showed that boosting methods outperformed logistic regression and Random Forests for high-dimensional EHR datasets. While their approach leveraged raw and engineered features, the heavy reliance on feature engineering meant they missed latent patterns not easily captured in this ML framework. In contrast, we included transformer-based models that learn sequence and context-aware representations from raw EHR features, without requiring extensive manual transformations.

Overall, these studies emphasized the superiority of non-linear models for clinical risk predictions, the value in feature engineering for treatment intensity and change variables, and the value of benchmarking multiple model types. Our work harmonizes these insights and applies a mixed use of logistic regression, Random Forest, XGBoost, and transformer-based neural networks to highlight the clinical relevance of the most impactful predictors.

Dataset

The dataset used in this paper originates from the University of California, Irvine (UCI) Machine Learning Repository (*Diabetes*), a well-known source for benchmark datasets in data science research. This dataset is widely studied in healthcare analytics as it provides valuable insights into risk factors and predictors of hospital readmission among diabetic patients. It contains 101,766 patient records collected between 1999 and 2008 across 130 different hospitals and integrated delivery networks in the United States. Each record corresponds to a patient diagnosed with diabetes who received laboratory tests, was prescribed medication, and experienced a hospital stay of up to 14 days. In total, the dataset provided 50 input features spanning both categorical and numerical variables. Our main focus was on the binary target variable, *readmission*, which indicates whether a patient was readmitted or was not readmitted to the hospital within 30 days of discharge.

Data Preprocessing

One of the most significant changes we took during preprocessing was the transformation of categorical data into one-hot encoded data. For example, the original dataset had 20+ columns, each representing a medication, containing information about whether the drug was prescribed or if there was a change in the dosage. Values for these columns included *up, down, steady,* or *no*. These values were transformed into four different dummy variables for each medication, increasing the total number of columns. Additionally, there were 8 continuous variables describing the prior number of hospital visits and medical history. These were all normalized using a min-max scaler.

Some columns were dropped due to their irrelevance to our prediction task. For example, *encounter_id* and *patient_id* were unique identifiers that provided no predictive power in our use case. The *weight* column was also dropped due to 95% of the values missing. We chose to keep one column with missing values, *race*. This is a categorical feature, and imputation logic was applied to create a ‘Missing’ value. Lastly, some patients had multiple encounters, creating duplicate rows in the dataset. The first instance of each patient was kept, and the additional rows were dropped as were solely focused on first-time hospital visits. At the end of preprocessing, 42 features and 71,518 samples remained, containing categorical information on medication and demographics, and quantitative information on healthcare utilization and treatment intensity - such as number of procedures, time in hospital, etc. However, due to the extreme 10:1 imbalance of our classes in the dataset, we decided to take an additional step of under-sampling, ensuring the final dataset for the model had a 1:1 ratio of observations between the readmission and non-readmission classes. This strategy reduced the observation count to 12,586 samples while the feature count was unchanged at 42.

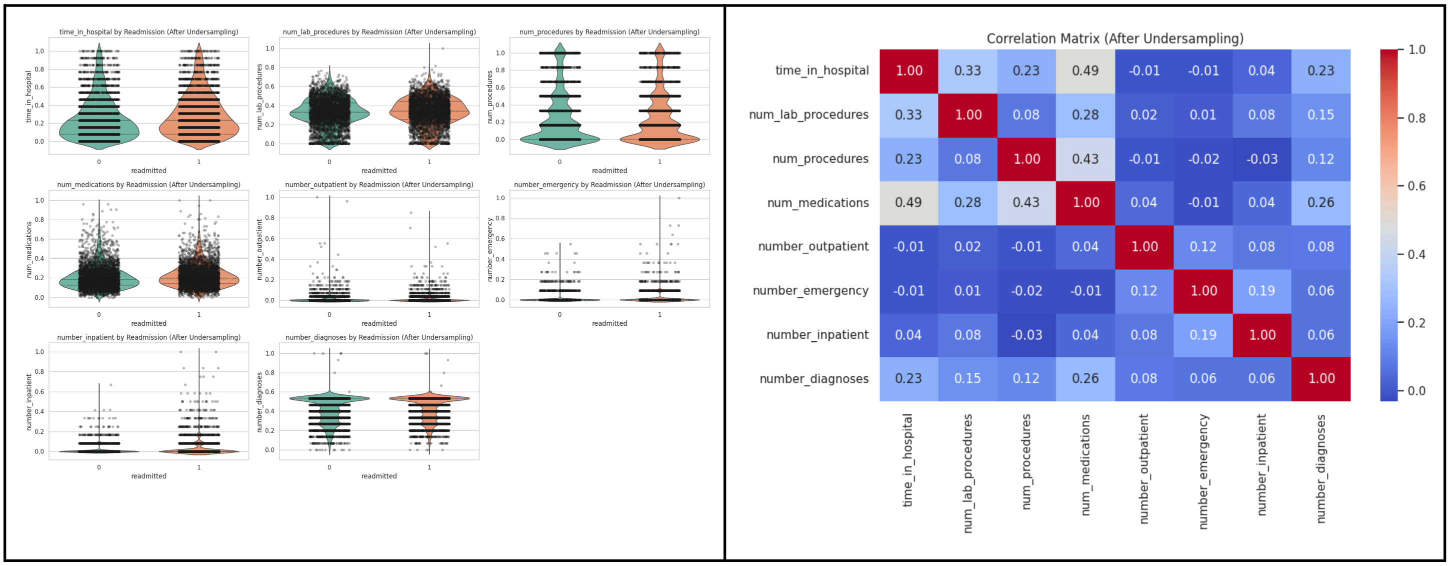
Dataset	Original Count	Undersampled Count	Split %
Training	42,910	7,551	60
Validation	14,304	2,517	20
Testing	14,304	2,518	20

Table 1: Comparison of dataset sizes before and after undersampling. The original dataset contained 42,910 training examples and 14,304 each for validation and testing. For model training, the data was undersampled to 7,551 training, 2,517 validation, and 2,518 testing examples, maintaining a 60/20/20 split.

Exploratory Data Analysis

Our exploratory data analysis provided a comprehensive understanding of the dataset’s structure, feature distributions, and relationships between the response variable and potential predictors. The continuous variables, including time_in_hospital, num_lab_procedures, num_procedures, num_medications, and number_diagnoses, presented right-skewed distributions, indicating that most patients had fewer medical interventions and shorter hospitalizations; However, a small subset of the population underwent substantially higher utilization, which revealed chronic or severe cases. Health utilization indicators, such as number_outpatient, number_emergency, and number_inpatient, were heavily concentrated near zero, suggesting that the majority of patients had limited hospital encounters beyond their primary admission. The categorical features, including race, gender, and age, showcased demographic imbalances, with older and Caucasian patients overrepresented in our dataset, which may reflect underlying disparities in those who have access to healthcare. The binary treatment indicators, change and diabetesMed, revealed that starting or adjusting diabetes medication was linked to higher readmission rates.

Next, we performed correlation analysis using Pearson coefficients, which revealed positive associations among hospital utilization features, confirming that patients with more inpatient visits also tended to have longer stays. In addition, a mutual information analysis identified num_medications, number_inpatient, and time_in_hospital as the strongest predictors of readmission, highlighting how treatment complexity and intensity of care play a key role in patient outcomes. Overall, these findings suggest that frequent hospital use, medication adjustments, and complex clinical profiles are key risk factors for readmission.



Methods
Our approach to evaluating a model's performance centered around the F1 score as a primary metric and accuracy as a secondary metric. We utilized undersampling during preprocessing to train all of our models on a balanced dataset, reaching a near 1:1 class ratio. F1 score was chosen above our metrics, as it provides a robust measure of discriminative power by weighting both precision and recall, which is critical for clinical intervention. The following models were implemented:

Logistic Regression (Baseline)
The first model that we implemented as the baseline was a Logistic Regression model. A logistic regression model works by determining the linear combination of initialized weights and inputs (along with a bias), and then passing this outcome through the Sigmoid activation function, to return outcomes within a standard (0,1) probability range. We applied a threshold of 0.5 to classify the outcome into one of the two categories. This was an appropriate model for our problem as the Logistic Regression classifier naturally predicts a binary outcome, which applies to the scenario of trying to determine if a patient is readmitted or not. All remaining hyperparameters were kept in their default setting.

Random Forest Classifier

The second model we use is a Random Forest classifier. The Random Forest model consists of building *k* decision trees and then taking a majority vote across the trees to determine the appropriate output. Random Forests are considered a form of "bagging" - an ensemble method where multiple models are independently trained on bootstrapped samples, and their outputs aggregated. This ensemble approach is effective, as averaging many decision trees tends to lower variance and reduce overfitting. This was an advantage when modeling high-dimensional EHR data. The Random Forest's ensemble design also allowed it to model complex non-linear relationships among medication changes, demographics, and hospitalization features, resulting in a more stable and discriminative decision boundary for this binary task.

XGBoost
The third classifier we employed was an XGBoost model. In juxtaposition to the aforementioned tree model, the XGBoost model builds a singular, strong model by sequentially combining many weak learners, or shallow decision trees. This consists of a technique called "boosting" - training additive models sequentially, where each one predicts the residual from the previous model. Boosting is

highly accurate and reduces both bias and variance. However, XGBoost models are more sensitive to noise and therefore more prone to overfitting compared to bagging. Similar to the Random Forest approach, XGBoost is effective for binary classification as it performs well when it comes to establishing a clear decision boundary, particularly when dealing with non-linear relationships. XGBoost also computes feature importance by measuring how much each variable improves split quality across the boosting process. These importance scores help reveal which clinical and demographic features meaningfully influence the model's output, supporting both interpretation and clinical relevance.

Transformer-based Neural Network Model
The fourth classifier we employed was a Transformer-based Neural Network specialized for a binary classification task. It processes a single vector of inputs through a dense layer to create an initial embedding, which is then reshaped to simulate a sequence of length one. We then passed this embedded representation through a single Transformer Block, which consists of a MultiHeadAttention layer followed by a Feed-Forward Network (FFN); residual connections and Layer Normalization are applied after both the attention and FFN sub-layers, which are standard components of the Transformer architecture. Finally, the output of the Transformer Block is flattened, passed through a final dense layer, which ends with a sigmoid output of the two classes we want to classify.

Experiments, Results, and Discussion
Drawing inspiration from Liu et al. 's 2024 comparison study, which evaluated 10 different models on the same dataset (albeit with a different outcome variable interpretation), we tested several different approaches of feature engineering and hyperparameter tuning on the RF and XG Boost models, which consistently performed the best across previous research.

Our primary experimental step involved addressing the significant 10:1 class imbalance present in the raw dataset. To mitigate bias and obtain true performance metrics, we undersampled the majority class to achieve a near 1:1 class balance for training and testing (Class 0: 51.07%, Class 1: 48.93%). While this decision decreased overall accuracy, it produced meaningful F1 Scores, confirming the models' ability to distinguish true readmission risk. We also implemented Group K-Fold Cross Validation on the patient number to ensure the same patient did not appear in both the training and validation sets, preventing data leakage.

Hyperparameter Turning and Validation Methods
For the Random Forest and XGBoost models, hyperparameters were optimized using randomized search techniques. Specifically, tree-based models employed RandomizedSearchCV to efficiently tune key parameters like tree depth, number of estimators, and learning rate, prioritizing the F1 score as the evaluation metric. The Logistic Regression baseline and the Transformer-based model utilized KerasTuner to optimize network architecture, including the number of layers, unit size, optimizer, and learning rate. The use of a dedicated, unchanged validation set ensured that hyperparameter selection was robust and did not contaminate the final performance assessment on the test set.

Results and Comparisons
Our final model performance on the balanced test set is summarized in Table 2 below:

Model/Classifier	Train Accuracy %	Test Accuracy %	F1 Score %	F1-Female %	F1-Male %
Logistic Regression (Baseline)	53.60	53.26	52.60	50.29	54.85
Random Forest	81.61	56.31	57.56	59.72	54.99
XGBoost	**60.35**	**56.47**	**57.91**	**59.81**	**55.76**
Transformer	55.94	52.70	46.04	NA	NA

Table 2: Model Performance on The Balanced Test Dataset

Our results confirmed our hypothesis that tree-based models and boosting methods (Random Forest and XGBoost) performed substantially better than both the Logistic Regression baseline and the Transformer-based model. The XGBoost model achieved the highest F1 Score (57.91%), confirming its strength in handling complex, high-dimensional data for binary classification. This outcome aligned with existing literature favoring gradient boosting for clinical risk prediction.

A clear performance gap was observed between the training accuracy and the testing accuracy for the tree-based models, particularly Random Forest (Training: 81.61%, Testing: 56.31%), indicating overfitting. This was anticipated due to the high-dimensional, noisy nature of EHR data. Mitigation strategies included restricting the complexity of the trees during hyperparameter tuning such as limiting maximum depth, and relying on the inherent regularization properties of the ensemble methods. The lower training-testing gap in the XGBoost model (Training: 60.35%, Testing: 56.47%) suggests better generalization and is a key reason for final selection.

Subgroup Analysis
In addition to our main analysis, we also conducted a final subgroup analysis on our baseline and tree-based models (Random Forest and XGBoost) to evaluate potential disparities across gender. As shown in Table 2, both the Random Forest and the XGBoost models demonstrated higher, yet equitable, predictive power for the female subgroup. Specifically, for our best-performing model, XGBoost, the F1 Score for the female subgroup was 59.81%, and for the male subgroup, it was slightly lower at 55.76%. Similarly, the Random

Forest model scored 59.72% for females and 54.99% for males. This observation suggests that, though not substantial, the predictive factors of these tree-based models are better at identifying diabetic cases in the female population.

Feature Importance Analysis

Feature contribution analysis revealed a difference in the model's risk-assessment approach. The Random Forest model prioritized patient acuity and clinical complexity (a high number of lab procedures). In contrast, our best model, XGBoost, identified prior healthcare utilization (number of inpatient visits) as the strongest predictor. Despite these differences, both the Random Forest and XGBoost models prioritized *time_in_hospital*, which measures the resource utilization during the index stay, as a critical predictor of subsequent readmission risk.

GWO / SMOTE / Group K-Fold CV

Although not used in our final results, we experimented with several prominent pre-modeling techniques in an attempt to optimize our model. *Feature selection* was carried out using the Grey Wolf Optimization (GWO) algorithm from the *Mealpy* package, which contains several cutting-edge meta-heuristic optimization algorithms. The GWO proposed by Mirjalili et al. in 2103 employs a swarm intelligence-based technique that is based on the social dominance hierarchy of grey wolves to find an optimal feature subset. To handle the notable *class imbalance* between admitted and re-admitted (10:1 imbalance ratio), Synthetic Minority Over-Sampling (SMOTE) was used, which synthetically resamples the minority class to match the majority class. Finally, to prevent *data leakage* from the same patients, we used a Group K-Fold on the patient_nbr group to ensure the same patients did not appear in both the trains and validation sets. Combining GWO and SMOTE with k-neighbors=5 applied to each of 5 CV folds showed promising performance on the majority class, but was a poor predictor of the minority class, leading therefore was chosen to be omitted from our final model.

Conclusions

This project applied four different machine learning models to predict the 30-day readmission rate for diabetic patients using electronic health records. Our initial results demonstrated the profound challenge of class imbalance, where models defaulted to the majority class and yielded misleadingly high accuracy with near-zero F1 Scores. To obtain true performance metrics, we undersampled the majority class and achieved a 1:1 class balance. On this data, XGBoost demonstrated the strongest overall performance, achieving the highest test accuracy of 56.47% and the highest F1 Score of 57.91%. The Random Forest model was closely competitive, reaching 56.31% test accuracy and 57.56% F1 Score, but this model suffered from overfitting. The Logistic Regression and Transformer-based model performed significantly worse; the Transformer-based model yielded the lowest F1 Score at 46.04%. This shift confirms the superior ability of gradient boosting methods to identify true readmission cases in a balanced setting.

Our findings highlighted the clinical utility of gradient boosting methods, such as XGBoost, which demonstrated a clear and meaningful ability to distinguish readmitted cases compared to others. Feature contribution analysis highlighted that the prior utilization of healthcare and resource utilization during the stay were the most influential predictors to readmission cases. A significant limitation of this study was the modest overall F1 Scores and the temporal boundary of the dataset (1999 to 2008). Even so, using a similar model, tuned to optimize F1 scores would assist hospitals to target high-risk patients and deliver necessary interventions, thereby improving CMS compliance and post-discharge outcomes. Future methods might involve integrating SHAP value analysis for deeper feature interpretation and conducting more extensive hyperparameter tuning on the Transformer architecture.

Contributions

- Colin Frishberg: Created visualizations, helped with EDA, Data Processing, and feature engineering experiments using SMOTE, GWO, and Group K-fold CV.
- Terra Jiang: Developed and implemented Multiclass Logistic Regression, Random Forest, and XGBoost models (model training, hyperparameter tuning, evaluations, and subgroup analysis). Completed the experiment and conclusion report sections. Conducted notebook separation and modularizations, including exporting models, model performance results, and train/val/test CSV files. Created the README file. Merged the terra_staging branch to the main. Finalized results, feature importance, and conclusion slides.
- Sai Sriya Mudigonda: Conducted exploratory data analysis, created visualizations for feature distributions, categorical trends, and correlations, and wrote abstract, updated introduction, related work, EDA, and part of experiments, results, and discussion section. Conducted feature engineering experiments with oversampling and SVD on LR and XGBoost.
- Rahil Sharma: Completed the data overview and data preprocessing sections - feature selection, feature engineering, shuffling, train/val/test split. Completed the undersampling and implemented a LR and XGBoost model. Worked on slides.

Project GitHub Repo

https://github.com/UC-Berkeley-I-School/w207_final_project_frishberg_jiang_mudigonda_sharma

References

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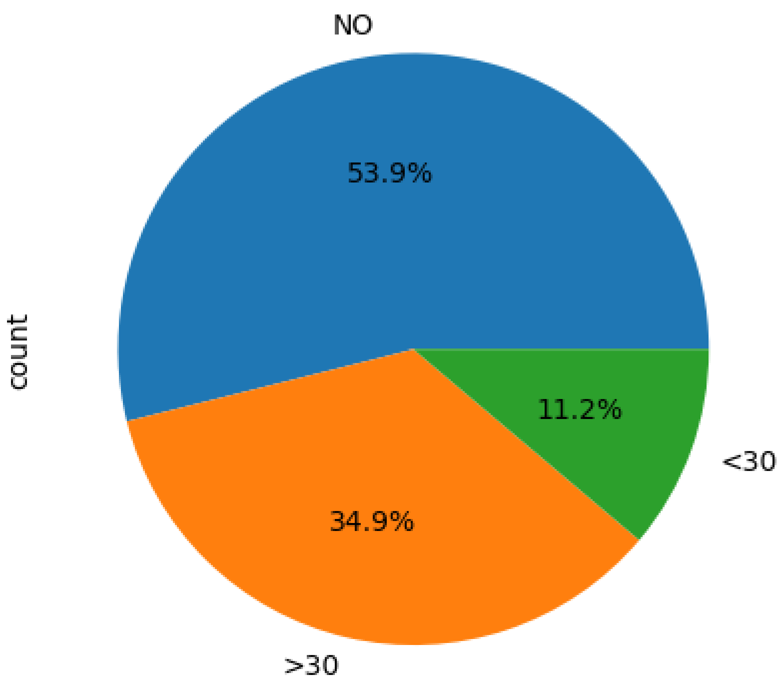
Liu V.B., L.Y. Sue, Y. Wu “Comparison of Machine Learning Models for Predicting 30-day Readmission Rates for Patients with Diabetes.” Journal of Medical Artificial Intelligence, vol. 7,2024, p.23.

Mirjalili, Seyedali, Seyed Mohammad Mirjalili, & Andrew Lewis. “Grey Wolf Optimizer.” ***Advances in Engineering Software***, vol. *69*, 2014, pp. 46–61.

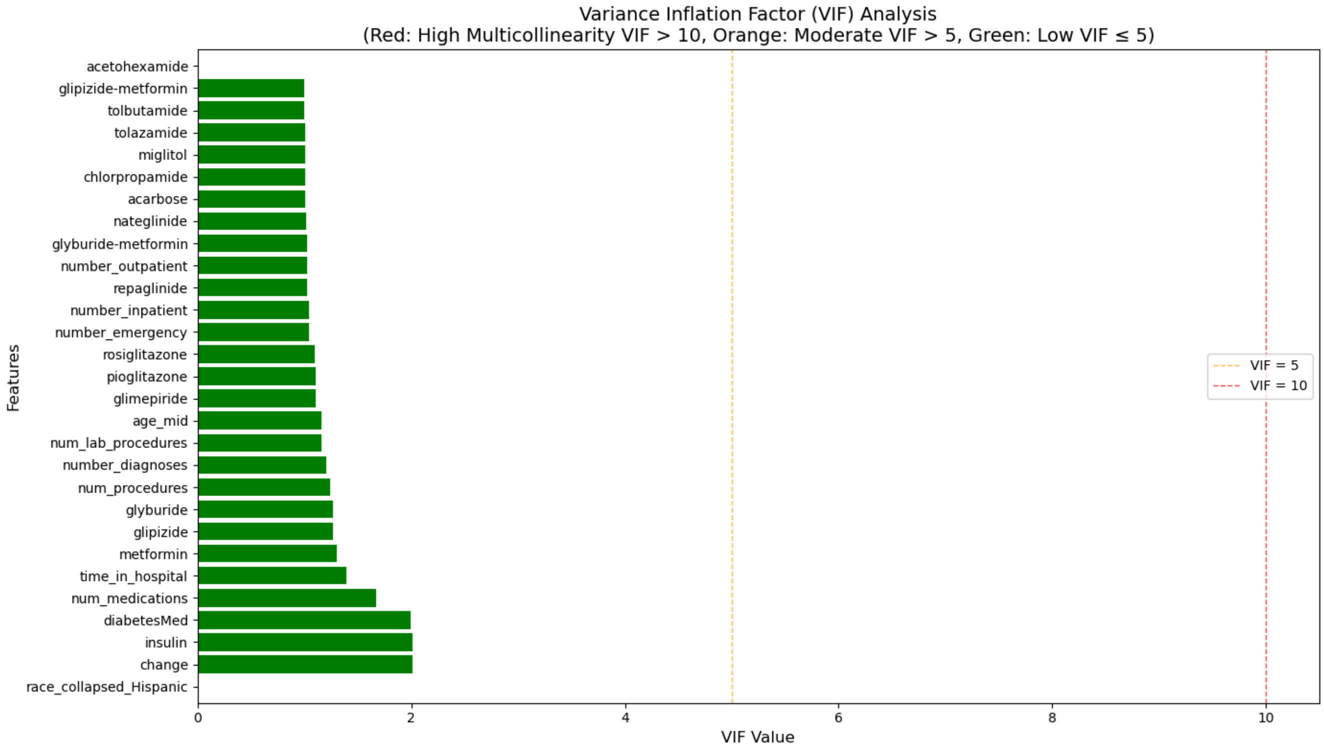
Yu, Shan, Farooq Farooq, Jing Hua, Aoluo Wang, and Greg Finley. “Predicting Readmission Risk for Patients with Diabetes Using Electronic Health Record Data.” Journal of Diabetes Science and Technology, vol.9, no. 6, 2015, pp. 1206-15.

Appendix

a)



b)


Variance Inflation Factor (VIF) Analysis
(Red: High Multicollinearity VIF > 10, Orange: Moderate VIF > 5, Green: Low VIF ≤ 5)
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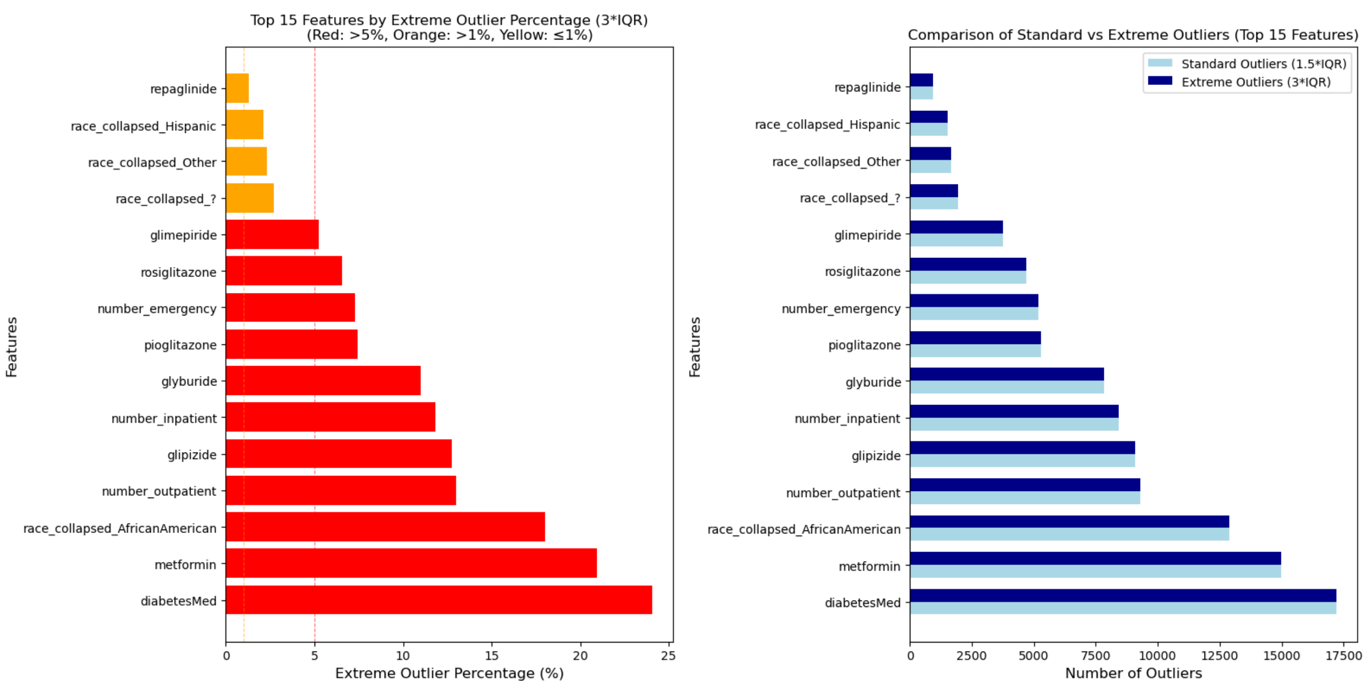
glipizide-metformin
tolbutamide
tolazamide
miglitol
chlorpropamide
acarbose
nateglinide
glyburide-metformin
number_outpatient
repaglinide
number_inpatient
number_emergency
rosiglitazone
pioglitazone
glimepiride
age_mid
num_lab_procedures
number_diagnoses
num_procedures
glyburide
glipizide
metformin
time_in_hospital
num_medications
diabetesMed
insulin
change
race_collapsed_Hispanic
Features
VIF = 5
VIF = 10
0
2
4
6
8
10
VIF Value

c)


Top 15 Features by Extreme Outlier Percentage (3*IQR)
(Red: >5%, Orange: >1%, Yellow: ≤1%)
repaglinide
race_collapsed_Hispanic
race_collapsed_Other
race_collapsed_?
glimepiride
rosiglitazone
number_emergency
pioglitazone
glyburide
number_inpatient
glipizide
number_outpatient
race_collapsed_AfricanAmerican
metformin
diabetesMed
Features
0
5
10
15
20
25
Extreme Outlier Percentage (%)
Comparison of Standard vs Extreme Outliers (Top 15 Features)
Standard Outliers (1.5*IQR)
Extreme Outliers (3*IQR)
repaglinide
race_collapsed_Hispanic
race_collapsed_Other
race_collapsed_?
glimepiride
rosiglitazone
number_emergency
pioglitazone
glyburide
number_inpatient
glipizide
number_outpatient
race_collapsed_AfricanAmerican
metformin
diabetesMed
Features
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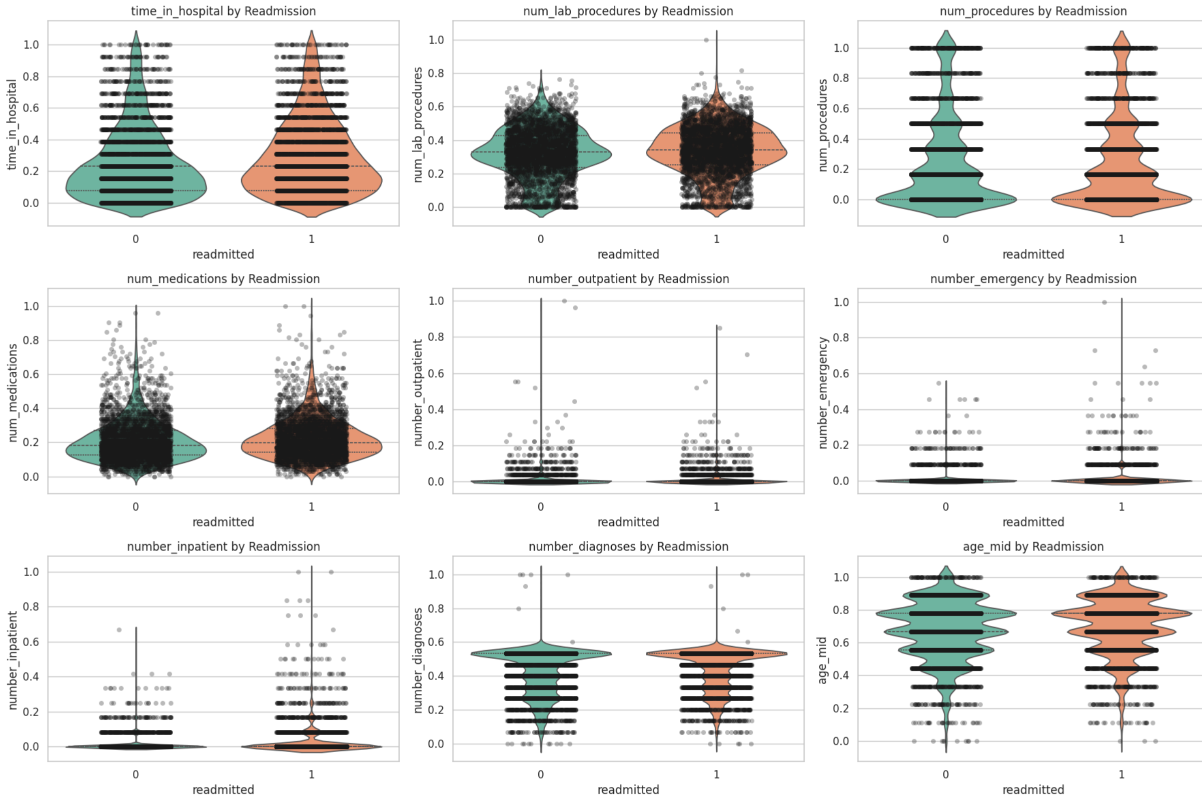
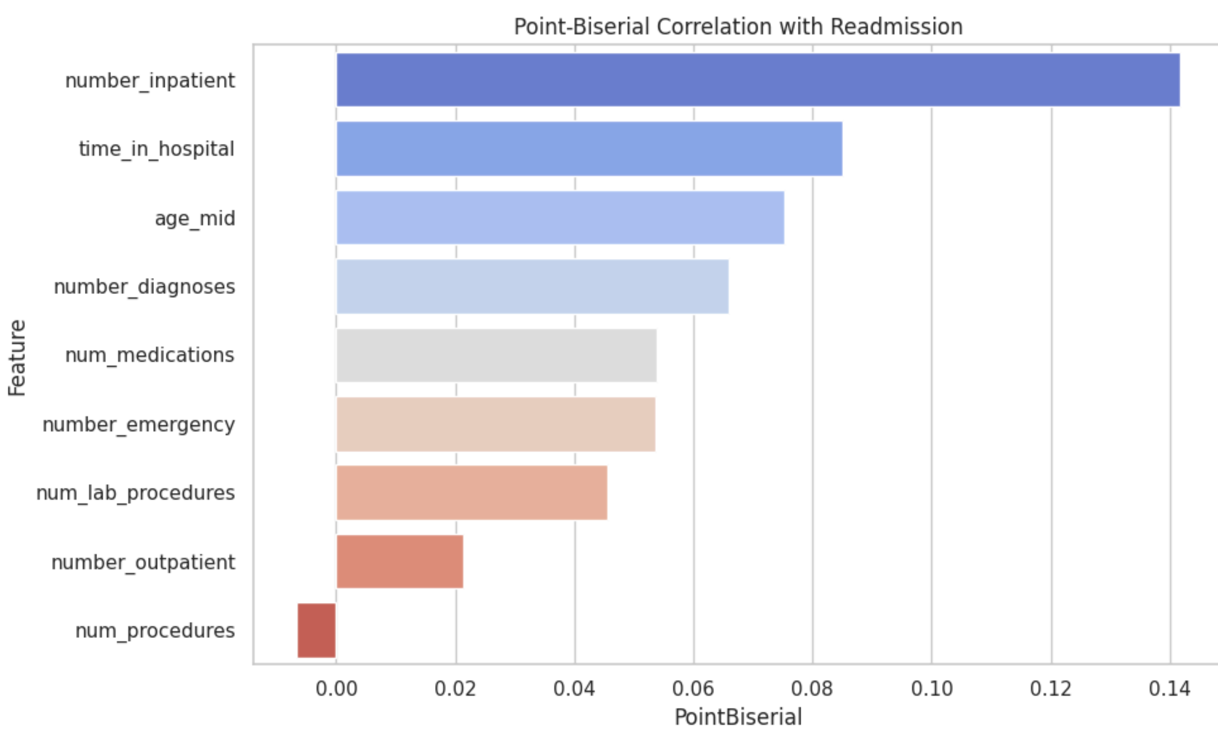
7500
10000
12500
15000
17500
Number of Outliers

d)


time_in_hospital by Readmission
num_lab_procedures by Readmission
num_procedures by Readmission
num_medications by Readmission
number_outpatient by Readmission
number_emergency by Readmission
number_inpatient by Readmission
number_diagnoses by Readmission
age_mid by Readmission
readmitted

e)


Point-Biserial Correlation with Readmission
number_inpatient
time_in_hospital
age_mid
number_diagnoses
num_medications
number_emergency
num_lab_procedures
number_outpatient
num_procedures
Feature
PointBiserial